THE FILTERED SPECTRAL SEQUENCE FOR A LEFT EXACT FUNCTOR ON FILTERED COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 8722-8773, following the statement at 8700-8720.

Editorial note. No omissions: every equality in the E_1 chain is displayed with its reason.

Allowed standard prerequisites (editorial).

- Filtered injective resolution. lemma-right-resolution-by-filtered-injectives (carded L26168), cited at proof lines 8725 and 8756. Supplies the bounded below filtered injective resolutions K -> I and L -> J.
- Spectral sequence of a filtered complex. homology-lemma-spectral-sequence-filtered-complex and its functorial variant, cross-chapter citations at proof lines 8733 and 8771 (tags mechanism). Supplies E_1^{p,q} = H^{p+q}(gr^p(T(I))) and the functorial morphism of spectral sequences.
- Biregular spectral sequences converge. homology-lemma-biregular-ss-converges, cross-chapter citation at proof line 8745 (tags mechanism). Supplies boundedness, convergence to R^n T(K) and the finite filtrations.
- Maps into filtered injective complexes. lemma-morphisms-into-filtered-injective-complex, cited at proof line 8759; complete proof at source lines 8482-8523, queued for separate carding. Supplies the homotopy-commutative square between the two resolutions used for functoriality.

Source reference. Original equation 015T: sources/stacks-derived/pinned-original-source.tex, source lines 8635–8635. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 05U2: sources/stacks-derived/pinned-original-source.tex, source lines 8659–8659. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 012W: focused-seeds/geometry/S9-der-ss-filtered-derived/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 012M: focused-seeds/geometry/S9-der-ss-filtered-derived/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 012O: focused-seeds/geometry/S9-der-ss-filtered-derived/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 012K: focused-seeds/geometry/S9-der-ss-filtered-derived/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TY: sources/stacks-derived/pinned-original-source.tex, source lines 8458–8458. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TW: sources/stacks-derived/pinned-original-source.tex, source lines 8313–8313. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories. Let $T : \mathcal{A} \to \mathcal{B}$ be a left exact functor. Assume $\mathcal{A}$ has enough injectives. Let $(K^\bullet, F)$ be an object of $Comp^+(Fil^f(\mathcal{A}))$. There exists a spectral sequence $(E_r, d_r)_{r\geq 0}$ consisting of bigraded objects E_r of $\mathcal{B}$ and d_r of bidegree $(r, -r+1)$ and with*

$$E_1^{p,q} = R^{p+q}T(gr^p(K^\bullet))$$

Moreover, this spectral sequence is bounded, converges to $R^*T(K^\bullet)$, and induces a finite filtration on each $R^nT(K^\bullet)$. The construction of this spectral sequence is functorial in the object $K^\bullet$ of $Comp^+(Fil^f(\mathcal{A}))$ and the terms (E_r, d_r) for $r \geq 1$ do not depend on any choices.

Proof. Choose a filtered quasi-isomorphism $K^\bullet \to I^\bullet$ with $I^\bullet$ a bounded below complex of filtered injective objects, see Lemma 05TW. Consider the complex $RT(K^\bullet) = T_{ext}(I^\bullet)$, see (05U2). Thus we can consider the spectral sequence $(E_r, d_r)_{r\geq 0}$ associated to this as a filtered complex in $\mathcal{B}$, see Homology, Section 012K. By Homology, Lemma 012M we have $E_1^{p,q} = H^{p+q}(\mathrm{gr}^p(T(I^\bullet)))$. By Equation (015T) we have $E_1^{p,q} = H^{p+q}(T(\mathrm{gr}^p(I^\bullet)))$, and by definition of a filtered injective resolution the map $\mathrm{gr}^p(K^\bullet) \to \mathrm{gr}^p(I^\bullet)$ is an injective resolution. Hence $E_1^{p,q} = R^{p+q}T(\mathrm{gr}^p(K^\bullet))$.

On the other hand, each I^n has a finite filtration and hence each $T(I^n)$ has a finite filtration. Thus we may apply Homology, Lemma 012W to conclude that

the spectral sequence is bounded, converges to $H^n(T(I^\bullet)) = R^nT(K^\bullet)$ moreover inducing finite filtrations on each of the terms.

Suppose that $K^\bullet \to L^\bullet$ is a morphism of $\mathrm{Comp}^+(\mathrm{Fil}^f(\mathcal{A}))$. Choose a filtered quasi-isomorphism $L^\bullet \to J^\bullet$ with $J^\bullet$ a bounded below complex of filtered injective objects, see Lemma 05TW. By our results above, for example Lemma 05TY, there exists a diagram

$$\begin{array}{ccc} K^\bullet & \longrightarrow & L^\bullet \\ \downarrow & & \downarrow \\ I^\bullet & \longrightarrow & J^\bullet \end{array}$$

which commutes up to homotopy. Hence we get a morphism of filtered complexes $T(I^\bullet) \to T(J^\bullet)$ which gives rise to the morphism of spectral sequences, see Homology, Lemma 012O. The last statement follows from this. □